Linear regression (running + interpreting)

CRISP R Mini-Course

2026

Day 6

Review from last time

What does the code below do? Explain to your neighbor!

```
Library(dplyr)

df <- read.csv("data.csv")

df_new <- df %>%
      mutate(sex = factor(sex, labels = c("Female", "Male")),
          disease = factor(disease, labels = c("no", "yes")) %>%
   filter(bmi < 30)

tx_disease_table <- table(df_new %>% select(sex, disease))

prop.test(tx_disease_table)
```

CRISP R course roadmap (1)

Step 0:

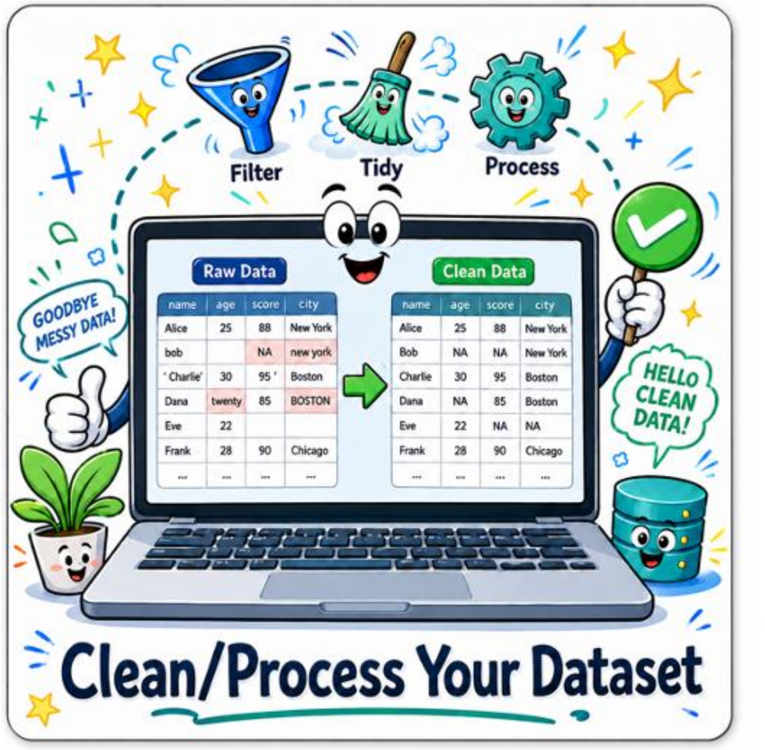
Setup R

Step 1:

Read in dataset

Step 2:

Clean/process your dataset



All images courtesy of ChatGPT

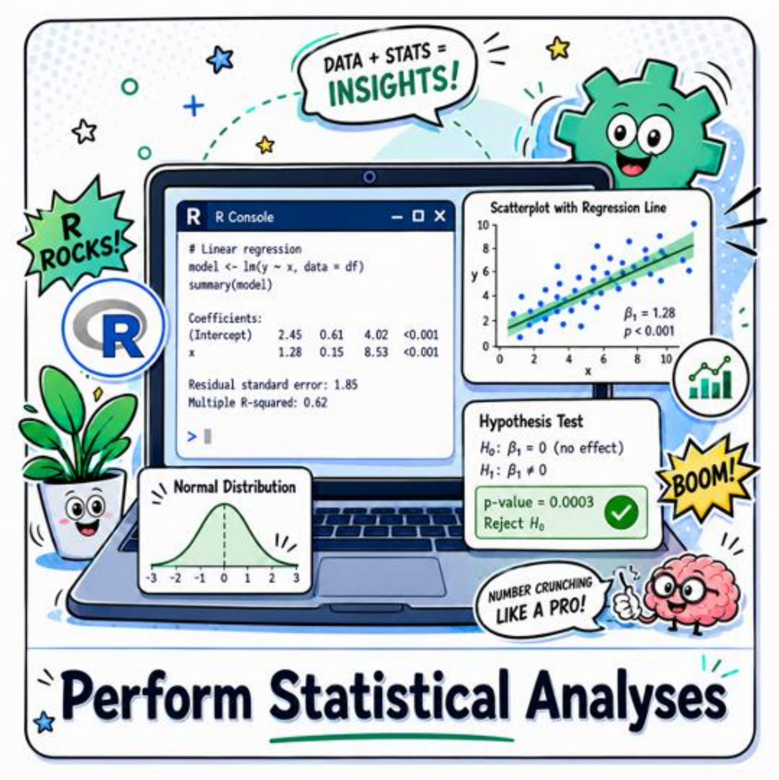
CRISP R course roadmap (2)

Step 3:

Obtain descriptive statistics

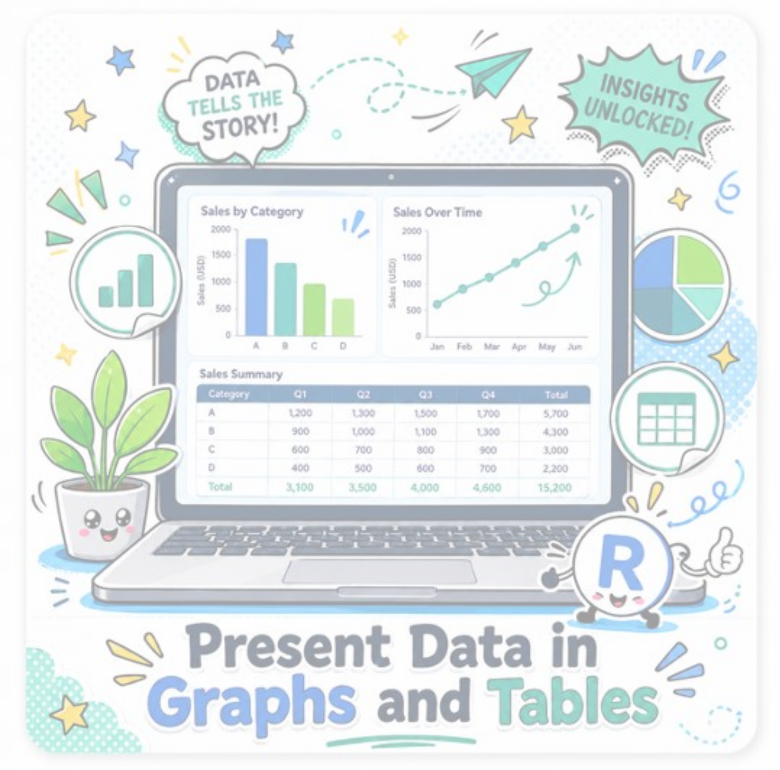
Step 4:

Perform statistical analyses



Step 5:

Present data in graphs and tables



Today’s agenda

- Linear regression – conceptual tutorial
- Running and interpreting linear regression in R

Guided tutorial

Today, we will learn the basics of linear regression.

1. Go to bit.ly/crisp2026.
2. Download (1) new CRISP survey data file, and (2) Rmd file for today into your CRISP R notes folder.

Linear regression w/ continuous variable (1)

- We are trying to determine if a years of education is associated with age. We obtain data from an observational study with both variables.
- We answer this question using linear regression:
  - **Outcome**: Years of education
  - **Exposure**: Age

Linear regression w/ continuous variable (2)

- Linear regression:
  - **Outcome**: Years of education
  - **Exposure**: Age

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.30130    0.77132  18.541  < 2e-16 ***
age         -0.06426    0.02275  -2.825  0.00573 ** 
```

Linear regression w/ binary variable (1)

- We are trying to determine if a number of years of education is associated with sex. We obtain data from an observational study with both variables.
- *Q: What is one way we can analyze our data to answer this question?*

Linear regression w/ binary variable (2)

- We are trying to determine if a number of years of education is associated with sex. We obtain data from an observational study with both variables.
- We can also analyze this using linear regression:
  - **Outcome**: Number of years of education
  - **Exposure**: Sex

Linear regression w/ binary variable (3)

- Linear regression:
  - **Outcome**: Years of education
  - **Exposure**: Sex

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.21127    0.25203  48.451   <2e-16 ***
sexfemale   -0.03885    0.46801  -0.083    0.934
```

Linear regression w/ adjustment variable (1)

- We are trying to determine if number of years of education is associated with age adjusting for sex. We obtain data from an observational study with all variables.
- We answer this question using linear regression:
  - **Outcome**: Number of years of education
  - **Exposure**: Age
  - **Adjustment covariate**: Sex

Linear regression w/ adjustment variable (2)

- Linear regression:
  - **Outcome**: Number of years of education
  - **Exposure**: Age
  - **Adjustment covariate**: Sex

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.36048    0.79808  17.994  < 2e-16 ***
age         -0.06482    0.02292  -2.828  0.00569 ** 
sexfemale   -0.14054    0.45358  -0.310  0.75733    
```